

HW-2: Linear regression and classification

This homework is based on exercises from ISLP-chapter 3 and chapter-4

<https://www.statlearning.com/>

Instructions: Please complete the following exercises using `python` in a single Jupyter notebook, export the results to HTML or PDF for submission to Canvas.

Note: You do not need to code algorithms from scratch, you can use StatsModels or any other package. However for this HW, **rely on StatsModels instead of SkLearn**, as its is better suited for rigorous statistical analysis.

Grading: Each sub-bullet below will have a rubric category for the TA to score and quantify the quality of your submission for this portion of the deliverable , e.g. on 3.4.a there will be a rubric category with something like '100%, 85%, 70%, 55%, 0%'

Chapter-3: Exercises

Conceptual: Complete with textual and/or pen and paper responses

3.4. I collect a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$.

- (a) Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.
- (b) Answer (a) using test rather than training RSS.
- (c) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.
- (d) Answer (c) using test rather than training RSS.

3.7. Prove that in simple linear regression with intercept, $R^2 = \text{Corr}(X, Y)^2$. Assume $\bar{x} = \bar{y} = 0$. Also show that $R^2 = \text{Corr}(Y, \hat{Y})^2$, where $\hat{Y}$ are the fitted values.

Applied: Complete with code

3.9. This question involves the use of multiple linear regression on the Auto data set.

- (a) Produce a scatterplot matrix which includes all of the variables in the data set.
- (b) Compute the matrix of correlations between the variables using the `DataFrame.corr()`

method.

- (c) Use the `sm.OLS()` function to perform a multiple linear regression with `mpg` as the response and all other variables except `name` as the predictors. Use the `summarize()` function to print the results. Comment on the output. For instance:
 - i. Is there a relationship between the predictors and the response? Use the `anova_lm()` function from `statsmodels` to answer this question.
 - ii. Which predictors appear to have a statistically significant relationship to the response?
 - iii. What does the coefficient for the `year` variable suggest?
- (d) Produce some of diagnostic plots of the linear regression fit as described in the lab. Comment on any problems you see with the fit. Do the residual plots suggest any unusually large outliers? Does the leverage plot identify any observations with unusually high leverage?
- (e) Fit some models with interactions as described in the lab. Do any interactions appear to be statistically significant?
- (f) Try a few different transformations of the variables, such as $\log(X)$, $\sqrt{X}$, X^2 . Comment on your findings.

3.10. This question should be answered using the `Carseats` data set.

- (a) Fit a multiple regression model to predict `Sales` using `Price`, `Urban`, and `US`.
- (b) Provide an interpretation of each coefficient in the model. Be careful-some of the variables in the model are qualitative!
- (c) Write out the model in equation form, being careful to handle the qualitative variables properly.
- (d) For which of the predictors can you reject the null hypothesis $H_0 : \beta_j = 0$?
- (e) On the basis of your response to the previous question, fit a smaller model that only uses the predictors for which there is evidence of association with the outcome.
- (f) How well do the models in (a) and (e) fit the data?
- (g) Using the model from (e), obtain 95% confidence intervals for the coefficient(s).
- (h) Is there evidence of outliers or high leverage observations in the model from (e)?

Chapter-4: Exercises

Conceptual: Complete with textual and/or pen and paper responses

4.5. We now examine the differences between LDA and QDA.

- (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?

- (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?
- (c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?
- (d) True or False: Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.

4.6. Suppose we collect data for a group of students in a statistics class with variables $X_1 =$ hours studied, $X_2 =$ undergrad GPA, and $Y =$ receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

- (a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.
- (b) How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

4.12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \text{orange} \mid X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)} \quad (1)$$

Your friend fits a logistic regression model to the same data using the softmax formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \text{orange} \mid X = x) = \frac{\exp(\hat{\alpha}_{\text{orange } 0} + \hat{\alpha}_{\text{orange } 1} x)}{\exp(\hat{\alpha}_{\text{orange } 0} + \hat{\alpha}_{\text{orange } 1} x) + \exp(\hat{\alpha}_{\text{apple } 0} + \hat{\alpha}_{\text{apple } 1} x)}. \quad (2)$$

- (a) What is the log odds of orange versus apple in your model?
- (b) What is the log odds of orange versus apple in your friend's model?
- (c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.
- (d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\text{orange } 0} = 1.2$, $\hat{\alpha}_{\text{orange } 1} = -2$, $\hat{\alpha}_{\text{apple } 0} = 3$, $\hat{\alpha}_{\text{apple } 1} = 0.6$. What are the coefficient estimates in your model?
- (e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

Applied: Complete with code

4.13. This question should be answered using the Weekly data set, which is part of the ISLP package. This data is similar in nature to the `smarket` data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.

- (a) Produce some numerical and graphical summaries of the Weekly data. Do there appear to be any patterns?
- (b) Use the full data set to perform a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?
- (c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.
- (d) Now fit the logistic regression model using a training data period from 1990 to 2008, with Lag2 as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).
- (e) Repeat (d) using LDA.
- (f) Repeat (d) using QDA.
- (g) Repeat (d) using KNN with $K = 1$.
- (h) Repeat (d) using naive Bayes.
- (i) Which of these methods appears to provide the best results on this data?
- (j) Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.